```
'Ke'Shawn Stafford
'Project 3
'02/20/2024

' {$STAMP BS2}
' {$PBASIC 2.5}

row       VAR Nib
column    VAR Nib
keypad1   VAR Word
keypad2   VAR Word
temp      VAR Nib
counter   VAR Byte
LCD       CON 14
last      VAR Byte

PAUSE 1000
SEROUT LCD, 84, [22,12]
SEROUT LCD, 84, [12]

DO
    GOSUB READkeypad1
    IF keypad1 <> keypad2 THEN

        GOSUB display
        PAUSE 50
    ENDIF
    keypad2 = keypad1
LOOP

READkeypad1:
    keypad1 = 0
    OUTL = %00000000
    DIRL = %00000000

    FOR row = 0 TO 3
        DIRB = %1111
        OUTB = %0000
        OUTA = 1 << Row
        DIRA = 1 << Row

        temp = 0
        FOR column = 0 TO 3
            INPUT (column + 4)
            temp = temp | (INB & (1 << column))
        NEXT

        keypad1 = keypad1 << 4 | (Temp REV 4)
    NEXT
RETURN

display:

    IF keypad1.BIT15 THEN
     IF counter > 39 THEN
         SEROUT LCD, 84, [12]
         PAUSE 20
         counter = 0
         ENDIF
        SEROUT LCD, 84, ["1 ",208,218,220] : counter = counter + 1
        last = "1"
    ENDIF
    IF keypad1.BIT14 THEN
     IF counter > 39 THEN
         SEROUT LCD, 84, [12]
         PAUSE 20
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     counter = 0
     ENDIF
    SEROUT LCD, 84, ["2 ",208,218,221] : counter = counter + 1
    last = "2"
ENDIF
IF keypad1.BIT13 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["3 ",208,218,222] : counter = counter + 1
    last = "3"
ENDIF
IF keypad1.BIT12 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["A ",208,218,223] : counter = counter + 1
    last = "A"
ENDIF
IF keypad1.BIT11 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["4 ",208,218,224] : counter = counter + 1
    last = "4"
ENDIF
IF keypad1.BIT10 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["5 ",208,218,225] : counter = counter + 1
    last = "5"
ENDIF
IF keypad1.BIT9 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["6 ",208,218,226] : counter = counter + 1
    last = "6"
ENDIF
IF keypad1.BIT8 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["B ",208,218,227] : counter = counter + 1
    last = "B"
ENDIF
IF keypad1.BIT7 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["7 ",208,218,228] : counter = counter + 1
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    last = "7"
ENDIF
IF keypad1.BIT6 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["8 ",208,218,229] : counter = counter + 1
    last = "8"
ENDIF
IF keypad1.BIT5 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["9 ",208,218,230] : counter = counter + 1
    last = "9"
ENDIF
IF keypad1.BIT4 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["C ",208,218,231] : counter = counter + 1
    last = "C"
ENDIF
IF keypad1.BIT3 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["* ",208,219,220] : counter = counter + 1
    last = "*"
ENDIF
IF keypad1.BIT2 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["0 ",208,219,221] : counter = counter + 1
    last = "0"
ENDIF
IF keypad1.BIT1 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["# ",208,219,222] : counter = counter + 1
    last = "#"
ENDIF
IF keypad1.BIT0 THEN
 IF counter > 39 THEN
     SEROUT LCD, 84, [12]
     PAUSE 20
     counter = 0
     ENDIF
    SEROUT LCD, 84, ["D ",208,219,223] : counter = counter + 1
    last = "D"
ENDIF
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